

Scored against its own target, the unadjusted comparison ranked best; against a declared target it ranked worst

A simulation study answering catalog problem EST-11

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Abstract

Background. An estimator’s error against a declared decision target is its error against its own implicit (native) target plus the difference between the two targets. Simulation studies that score each method against its native target see only the first part. Catalog problem EST-11 asks whether that changes which method is recommended.

Methods. An anchored binary-outcome comparison with individual data on A versus C and a published B versus C trial; declared targets at three covariate means; B sharing A’s effect modification or not. Bucher, MAIC and STC (native target: the B trial’s population) and a two-stage STC carried to the declared target. The target mismatch was computed exactly; 1000 replicates per cell.

Results. The best method under the native metric differed from the best under the declared metric in 5 of 6 cells. The unadjusted Bucher comparison had the smallest error against the quantity it estimates, because it has the least variance, and a bias of 0.044 to 0.253 against the declared targets. When B’s modification differed from A’s, methods native to the B trial’s population missed a declared target one covariate SD away by 0.109, and the two-stage transport, which assumed shared modification, did not remove it.

Conclusion. Method comparisons should score every method against one declared target. Scoring each against its own target rewards the estimator that answers the wrong question most precisely.

1 The problem

$\hat{\Delta}_E - \Delta(F_D) = \{\hat{\Delta}_E - \Delta(F_E)\} + \{\Delta(F_E) - \Delta(F_D)\}$. The second term is fixed by the populations and the modification structure, does not shrink with sample size, and is zero by construction when a method is scored against its native target. A second-stage transport maps F_E to F_D ; it removes the term if its assumptions hold and replaces it otherwise. Anchored population adjustment is typically scored in the aggregate trial’s population (1).

2 Design

Registered protocol: `protocol.md`. $\text{logit } p = -0.5 + 0.5x + a_A(-0.5 + 0.3x) + a_B(-0.7 + g_Bx)$; individual data on A versus C with $x \sim N(0, 1)$, B versus C published from $x \sim N(0.6, 1)$; 300 per arm. $g_B = 0.3$ (B shares A’s modification) or 0.6. Declared targets $x \sim N(m_D, 1)$ with $m_D = 0, 0.6, 1.2$. The Bucher comparison’s native target mixes the two trial populations; MAIC and STC target the B trial’s population; the two-stage STC calibrates B’s effect to the published log odds ratio under shared modification and carries both contrasts to F_D .

3 Results

Table 1: Lowest RMSE per cell under each scoring rule; 1000 replicates.

declared target mean	B's modification	best by native RMSE	best by declared RMSE
0.0	0.3	bucher	stc
0.6	0.3	bucher	stc
1.2	0.3	bucher	stc_2stage
0.0	0.6	bucher	stc
0.6	0.6	bucher	stc_2stage
1.2	0.6	bucher	bucher

Table 2: MCSE of bias about 0.008.

declared target mean	B's modification	method	mismatch (exact)	RMSE native	RMSE declared	bias declared
0.0	0.3	bucher	0.143	0.241	0.279	0.141
0.0	0.3	maic	0.002	0.274	0.274	0.001
0.0	0.3	stc	0.002	0.261	0.261	0.006
0.0	0.3	stc_2stage	0.000	0.265	0.265	0.003
0.6	0.3	bucher	0.141	0.242	0.275	0.130
0.6	0.3	maic	0.000	0.265	0.265	-0.009
0.6	0.3	stc	0.000	0.251	0.251	-0.010
0.6	0.3	stc_2stage	0.000	0.251	0.251	-0.010
1.2	0.3	maic	-0.001	0.266	0.266	-0.001
1.2	0.3	bucher	0.140	0.242	0.282	0.145
1.2	0.3	stc_2stage	0.000	0.252	0.252	0.001
1.2	0.3	stc	-0.001	0.253	0.253	-0.000
0.0	0.6	maic	0.113	0.272	0.294	0.112
0.0	0.6	bucher	0.254	0.246	0.353	0.253
0.0	0.6	stc_2stage	0.000	0.281	0.281	0.111
0.0	0.6	stc	0.113	0.255	0.278	0.111
0.6	0.6	bucher	0.141	0.241	0.284	0.150
0.6	0.6	maic	0.000	0.275	0.275	0.005
0.6	0.6	stc	0.000	0.254	0.254	0.005
0.6	0.6	stc_2stage	0.000	0.254	0.254	0.005
1.2	0.6	bucher	0.032	0.248	0.251	0.044
1.2	0.6	maic	-0.109	0.272	0.288	-0.096
1.2	0.6	stc	-0.109	0.259	0.277	-0.099
1.2	0.6	stc_2stage	0.000	0.276	0.276	-0.098

The registered primary was confirmed (Table 1), and one mechanism drove it: the unadjusted comparison, scored against the mixture of populations it actually estimates, had the lowest RMSE in every cell, and against the declared target it was biased by 0.13 to 0.25 in five of six cells (Table 2). Its one declared-metric win, at the farthest target with unshared modification, was a coincidence of its mixed target lying near that target.

Among the adjusted methods, RMSE differences were within 0.03 and dominated by a sampling SD of about 0.25, so their order under either metric is not informative at this sample size. The mismatch term itself was large where the design predicted: with B's modification unlike A's, MAIC and STC were off by 0.11 at declared targets one covariate SD from the B trial's population. The two-stage transport removed the

mismatch when modification was shared and not when it was not, where its bias matched MAIC's; with modification shared, every adjusted method's target was already close to every declared target because the B-versus-A contrast had no conditional modification.

4 What this does not answer

Three declared targets, one covariate, one sample size, exact target moments, four methods; no ML-NMR or ML-UMR arm. Rankings among the adjusted methods would need larger trials or more replicates to resolve. Peer review has not been done.

References

1. David M. Phillippo, Sofia Dias, A. E. Ades, Mark Belger, Alan Brnabic, Alexander Schacht, Daniel Saure, Zbigniew Kadziola, Nicky J. Welton. Multilevel network meta-regression for population-adjusted treatment comparisons. *Journal of the Royal Statistical Society Series A*. 2020;183(3):1189–210. doi:[10.1111/rssa.12579](https://doi.org/10.1111/rssa.12579)